

## Submission STAT1-MID-SUB08

### Question STAT1-MID-Q01

scan faint / margin cut off

visible work:

$$\bar{x} = 25/5 = 5.00 \dots ?$$

last line is hard to read; maybe mean=5.00; sample variance=14.00

### Question STAT1-MID-Q02

scan faint / margin cut off

visible work:

$$P(D_1 \cap D_2) = (5/16)((5-1)/(16-1)) \dots ?$$

last line is hard to read; maybe 0.0833

### Question STAT1-MID-Q03

scan faint / margin cut off

visible work:

$$SE = \sqrt{(p(1-p)/n)} = \sqrt{(0.55(1-0.55)/40)} = 0.0787 \dots ?$$

last line is hard to read; maybe (0.396, 0.704); repeated-sampling interpretation

### Question STAT1-MID-Q04

scan faint / margin cut off

visible work:

Increase sample size n to reduce standard error.... ?

last line is hard to read; maybe increase n; lower confidence level or variability; note the coverage trade-off

### Question STAT1-MID-Q05

scan faint / margin cut off

visible work:

$0.03 < 0.05$ , so reject  $H_0$  under the stated rule.... ?

last line is hard to read; maybe reject  $H_0$ ; evidence against  $H_0$ , not proof that  $H_0$  is false

### Question STAT1-MID-Q06

scan faint / margin cut off

visible work:

$$\bar{x} = 35/5 = 7.00 \dots ?$$

last line is hard to read; maybe mean=7.00; sample variance=4.50

### Question STAT1-MID-Q07

scan faint / margin cut off

visible work:

$$P(D_1 \cap D_2) = (5/19)((5-1)/(19-1)) \dots ?$$

last line is hard to read; maybe 0.0585

Question STAT1-MID-Q08

scan faint / margin cut off

visible work:

$$SE = \sqrt{(p(1-p)/n)} = \sqrt{(0.35(1-0.35)/40)} = 0.0754... ?$$

last line is hard to read; maybe (0.202, 0.498); repeated-sampling interpretation